

Week 6 Deliverables

Traffic Congestion Prediction — AutoResearch
STAT 390 Capstone | Sherry Huang | 2026

New best result (exp_025): F1 = 0.6780 | Precision = 0.638 | Recall = 0.724
Model: LightGBM, n=300, depth=6, lr=0.05, num_leaves=63, undersample 2:1, threshold=0.40, 17 features

vs baseline (exp_001): F1 +0.1182 (+21.1%) | Recall +0.2793 (+62.8%) | 25 experiments, 0 crashes

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1 Revised Project Statement

What This Project Does

This project builds and iteratively improves a machine learning system that predicts whether a specific Chicago road segment will be **congested 30 minutes from now**, using only speed history and time-of-day features. No external data, no real-time feeds. The system runs as an AutoResearch loop: propose a change, run it, keep if F1 improves, revert if not.

What the Project Has Actually Demonstrated

The main contribution is identifying and partially fixing a structural labeling problem.

The Chicago traffic data uses a global congestion threshold (~23 mph, 30th percentile of all training speeds). This threshold does not account for the fact that different road segments have very different normal speeds. As a result, **~16% of training labels are incorrect**: highways at their normal slow pace are labeled congested, and fast side streets are not.

The fix — adding segment-relative features (per-segment mean speed, std, z-score, ratio) computed from training data only — broke through the F1 ceiling at 0.658 that no amount of model or hyperparameter tuning had been able to cross.

Stage	Best F1	What drove it
Baseline (exp_001)	0.5598	Logistic regression, lags + time only
Model selection (exp_003–007)	0.6566	Random Forest + extended lag features
Imbalance handling (exp_013–018)	0.6585	Undersampling 2:1 — ceiling reached
Segment features (exp_019–021)	0.6727	speed_zscore, speed_vs_seg_mean, segment_mean — ceiling broken
LightGBM + all features (exp_025)	0.6780	LightGBM edges RF; full 17-feature set

What This Project Is NOT Doing

- Not redefining the congestion label (run.py is frozen — label stays global)
- Not using external data, real-time feeds, or test data
- Not claiming to solve Chicago traffic — predicting congestion 30 min ahead on held-out validation data with a reproducible time-based split

The one-sentence claim:

Segment-relative speed features, combined with LightGBM and 2:1 undersampling, raise 30-minute-ahead congestion prediction F1 from 0.560 to 0.678 on Chicago Traffic Tracker data — breaking a labeling-driven ceiling that hyperparameter tuning alone could not cross.

2 Updated Agent Strategy (program.md)

Scope Lock — Week 6

Current best: LightGBM, n_estimators=300, max_depth=6, lr=0.05, num_leaves=63, undersample 2:1, threshold=0.40, 17 features. F1=0.6780, P=0.638, R=0.724.

Locked Search Space (Week 7 only)

Only LightGBM hyperparameters are eligible for further tuning. Budget is 5 experiments maximum. Decision threshold stays at 0.40.

Parameter	Current	Values to try
num_leaves	63	31, 127
min_child_samples	20 (default)	10, 50
colsample_bytree	1.0 (default)	0.8
subsample	1.0 (default)	0.8

Officially Dropped Directions

Direction	Reason dropped
HistGradientBoosting	Tried exp_005 (no seg features) and exp_022 (with seg features) — consistently below RF and LightGBM
XGBoost	exp_023 scored F1=0.6712 — competitive but loses to LightGBM on same setup
Threshold below 0.40	Raises recall but always drops F1; tried exp_015, 016, 018
New model families	Scope is closed per Week 6 lock
Redefining congestion label	Requires changing frozen run.py — out of scope
New features beyond current 17	Full available superset is now used

3 Ablation and Comparison Table

3a Model Comparison (same segment features + undersample 2:1 + threshold=0.40)

All four model families tested head-to-head with identical setup. LightGBM wins on F1 and trains fastest.

exp	Model	F1	Prec	Recall	vs RF	Status
exp_02_0	Random Forest n=200 depth=8	0.6727	0.644	0.704	—	prev best
exp_02_2	HistGradientBoosting max_iter=300 depth=6 lr=0.05	0.6553	0.540	0.834	−0.017	discard
exp_02_3	XGBoost n=300 depth=6 lr=0.05	0.6712	0.625	0.724	−0.002	discard
exp_02_4	LightGBM n=300 depth=6 lr=0.05 leaves=63	0.6736	0.627	0.727	+0.001	keep
exp_02_5	LightGBM — same + 3 extra features	0.6780	0.638	0.724	+0.005	best

3b Feature Ablation (on LightGBM)

Configuration	Features	F1	Delta
LightGBM, core 14 features (exp_024)	SPEED, lag_1-6, rolling_mean_3, speed_diff, HOUR, DAY_OF_WEEK, speed_zscore, speed_vs_seg_mean, segment_mean_speed	0.6736	—
+ rolling_std_3 + MONTH + segment_std_speed (exp_025)	All 17 available features	0.6780	+0.0044

The three added features each capture a different kind of road-specific variability: **rolling_std_3** (speed volatility over last 3 steps), **segment_std_speed** (how inherently noisy this road is), **MONTH** (Chicago seasonal patterns).

3c Full Experiment History — Summary by Direction

Experiments	Direction	F1 range	Verdict
exp_001–002	Logistic Regression baseline + balanced	0.560–0.636	LR ceiling found
exp_003–007	Random Forest model selection + n_estimators	0.645–0.657	RF wins over LR
exp_008–011	RF depth + feature set sweep	0.645–0.656	Tuning ceiling ~0.658
exp_012	RF no class weighting	0.623	Class balance essential — discard
exp_013–018	Undersampling ratios + threshold sweep	0.623–0.659	Cannot break 0.659 via threshold
exp_019–021	Segment-relative features added	0.664–0.673	Ceiling broken — key contribution

exp_022	HGB retry with segment features	0.655	Still loses to RF — discard
exp_023	XGBoost with segment features	0.671	Close but loses — discard
exp_024–025	LightGBM (14 then 17 features)	0.674–0.678	New best — locked

4 Locked Final Two-Week Plan

Story locked: LightGBM + segment-relative features broke the global-threshold ceiling. Weeks 7–8 refine and present this story — no new directions, no new model families.

The One Open Question

Can LightGBM hyperparameter tuning push F1 above 0.680? If yes: report the specific parameter and the gain. If no: F1=0.678 is the final result — that is still a complete and defensible story.

Week 7 — Hyperparameter Refinement Within LightGBM

Budget: 5 experiments max. Decision threshold stays at 0.40. Keep if F1 > 0.6780, discard otherwise. No exceptions.

exp	What to try	Rationale
exp_026	num_leaves=31 (smaller, less overfit)	Current 63 may be overfitting on 32k rows
exp_027	num_leaves=127 (more expressive)	More leaves = finer splits on segment patterns
exp_028	min_child_samples=10 (allow smaller leaves)	Segments with few samples may be underfit
exp_029	colsample_bytree=0.8 + subsample=0.8	Regularization via feature/row subsampling
exp_030	Best winner from above, n_estimators=500	More trees if best HP shows further room

Week 8 — Final Presentation and Submission

No new experiments. All effort goes to producing the final deliverable.

Task	Output
Final results summary	Updated results.csv + F1 trajectory plot across all runs
Feature importance analysis	LightGBM feature importance bar chart — which features matter most
Error analysis	What types of congestion events the model still misses (time of day, segment type)
Presentation slides	Narrative: problem → ceiling discovery → fix → model comparison → results
Week 8 deliverable PDF	All required sections including test set evaluation (first time test is used)

Locked Presentation Narrative

1	The prediction task and why it is hard 30-min-ahead congestion, class imbalance 3.5:1, global threshold issue
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2	The AutoResearch loop What it does, how it keeps experiments honest, why frozen run.py matters
3	The ceiling discovery Global threshold mislabels 16% of rows — no model can fix wrong labels
4	The fix Segment-relative features: what they are, how computed, why no leakage
5	Model comparison RF vs HGB vs XGB vs LightGBM — why LightGBM wins
6	Final result F1=0.678, +21% over baseline, +63% recall — with test set number
7	Remaining limitation Label definition still global — what a future project would change